

INFORMATION SYSTEMS MANAGEMENT

EXAM NOTES

Tech Management • Semester VIII

Ch 1 • Ch 2 • Ch 3 • Ch 5 • Ch 6 • Ch 8 • Ch 9 • Ch 10 • Ch 11 • Ch 12 • Ch 13

Based on Laudon & Laudon — MIS 17th Edition

TABLE OF CONTENTS

- Session 1** — Information Systems in Global Business Today
- Session 2 / Ch 2** — Global E-Business and Collaboration
- Chapter 3** — Information Systems, Organizations & Strategy
- Chapter 5** — IT Infrastructure & Emerging Technologies
- Chapter 6** — Foundations of Business Intelligence: Databases
- Chapter 8** — Securing Information Systems
- Chapter 9** — Enterprise Applications (ERP, SCM, CRM)
- Chapter 10** — E-Commerce: Digital Markets, Digital Goods
- Chapter 11** — Managing Knowledge & Artificial Intelligence
- Chapter 12** — Enhancing Decision Making
- Chapter 13** — Building Information Systems

Session 1

Information Systems in Global Business Today

1.1 What Is Information Systems Management (ISM)?

ISM is the **strategic study of how people, technology, and organisations interact.**

- **Goal:** Align a company's digital infrastructure with its business survival & growth.
- **Bridge Role:** ISM translates between business leaders (CEO/CFO) and technical specialists.
- **The Shift:** From "Digital Transformation" (building) → "Digital Excellence" (optimising for real-time performance).

1.2 Key Definitions

Term	Definition
Information Technology (IT)	Hardware & software a business uses
Information System (IS)	Interrelated components that collect, process, store & distribute information to support decision making
Data	Raw, unprocessed facts & figures
Information	Data shaped into meaningful, useful form
Digital Firm	Organisation where significant business relationships, core processes & assets are digitally mediated / managed

1.3 Functions of an IS (IPO Model)

INPUT (Data collection)	PROCESSING (Transformation)	OUTPUT (Information/Action)
-------------------------	-----------------------------	-----------------------------

1.4 Dimensions of Information Systems

Dimension	Key Elements	Example
Organisations	Hierarchies, business processes, culture, politics	Org chart, SOPs
Management	Planning, coordinating, deciding, controlling	KPIs, dashboards
Technology	Hardware, software, data mgmt, networking & telecoms	ERP, Cloud, IoT

1.5 Six Strategic Business Objectives of IS

ON-DISC (mnemonic)

O – Operational Excellence | N – New Products/Services/Models
 D – Decision Making (Improved) | I – Intimacy (Customer & Supplier)
 S – Survival | C – Competitive Advantage

Objective	What It Means	Indian / Global Example
Operational Excellence	Efficiency, productivity, profitability	Walmart RFID; UPS ORION routes
New Products / Models	Create new goods/services/business models	Apple iTunes; Blinkit dark stores
Customer & Supplier Intimacy	Serve customers well; lower supplier costs	Reliance Jio JPP; JC Penney–TAL VMI
Improved Decision Making	Real-time data replaces guesswork	Verizon dashboard; Jio proactive monitoring
Competitive Advantage	Out-perform rivals via IS	Amazon logistics; Walmart supply chain

Objective	What It Means	Indian / Global Example
Survival	Comply with mandates or keep up with rivals	Citibank ATMs; Sarbanes-Oxley compliance

1.6 What's New in MIS?

IT Innovations

- Cloud computing, Big Data, IoT
- Mobile digital platform
- AI & Machine Learning
- Social networks for business

Business & Market Changes

- E-commerce → \$3.6 trillion globally (2019)
- Mobile e-comm growing >20% p.a.
- Managers more mobile; use BI applications
- Flatter orgs; emphasis on competencies & data-driven decisions

1.7 The Digital Firm

Feature	Digital Firm Characteristic
Business Relationships	Digitally enabled (app/API/web portal) — no manual paperwork
Core Processes	Live in digital networks (e.g., Uber order-to-cash loop)
Corporate Assets	Managed digitally (RFID, cloud, AI risk management)
Flexibility	Time shifting (24/7) & Space shifting (remote/global)

1.8 Approaches to IS

- **Technical Approach:** Emphasises mathematical models & computing.
- **Behavioural Approach:** Focuses on development, maintenance, & human issues.
- **SocioTechnical Approach (Our Approach):** Optimise both social AND technical systems jointly. Technology alone ≠ productivity.

■ **Exam Tip:** The SocioTechnical view is a favourite exam point — technology investment alone does not guarantee productivity gains.

Session 2 / Chapter 2

Global E-Business and Collaboration

2.1 Business Processes

Business Process: A logically related set of tasks & activities required to produce a product or service.

- Firms are collections of business processes.
- **Functional processes:** Within a single department (e.g., payroll, quality check).
- **Cross-functional processes:** Span multiple departments (e.g., Order Fulfilment: Sales → Warehouse → Shipping → Finance).

How IT Improves Processes

- Automates manual steps (efficiency)
- Changes information flow — parallel instead of sequential steps
- Eliminates delays; enables new business models

2.2 IS for Different Management Groups

System Type	Who Uses It / Purpose
Transaction Processing System (TPS)	Operational staff — records daily routine transactions (sales orders, payroll, shipping)
Management Information System (MIS)	Middle managers — structured reports, monitoring operations
Decision Support System (DSS)	Middle & senior managers — semi-structured decisions; "what-if" analysis
Executive Support System (ESS)	C-level — non-routine decisions; dashboards, external data, drill-down
Business Intelligence (BI)	All levels — analytics, OLAP, data mining, scorecards

2.3 Enterprise Applications

Systems that span functional areas and execute business processes across the entire firm.

Application	Abbreviation	Core Purpose
Enterprise Resource Planning	ERP	Integrates data from all key business processes into one system
Supply Chain Management	SCM	Manages supplier, distributor & logistics relationships and info
Customer Relationship Management	CRM	Manages customer interactions across sales, marketing & service
Knowledge Management Systems	KMS	Captures and distributes knowledge and expertise firm-wide

2.4 E-Business, E-Commerce & E-Government

- **E-Business:** Use of digital technology & Internet to drive major business processes.
- **E-Commerce:** Subset of e-business — buying & selling via Internet.
- **E-Government:** Using Internet to deliver info and services to citizens, employees & businesses.

2.5 Intranets & Extranets

- **Intranet:** Internal private network using Internet standards — access restricted to employees.
- **Extranet:** Company website accessible to authorised vendors / suppliers — facilitates collaboration.

2.6 Collaboration & Social Business

- **Collaboration:** Short/long-term, informal/formal teamwork that drives productivity, quality & innovation.
- **Social Business:** Uses social networking platforms (internal & external) to deepen interactions and speed information sharing.

Tools for Collaboration

- Email & IM
- Wikis & Blogs
- Virtual meeting / videoconferencing / telepresence
- Cloud collaboration: Google Drive, Docs
- Microsoft SharePoint, IBM Notes
- Enterprise social networking: Slack, Yammer, Chatter

2.7 The IS Department

Role	Responsibility
CIO (Chief Information Officer)	Heads IS department; aligns IT with strategy
CSO (Chief Security Officer)	Manages information security
CKO (Chief Knowledge Officer)	Manages organisational knowledge assets
CPO (Chief Privacy Officer)	Ensures privacy compliance
CDO (Chief Data Officer)	Manages data assets and governance
■ Exam Tip: Know all C-suite IS roles and the difference between Intranet (internal) vs Extranet (suppliers/vendors).	

Chapter #3

Information Systems, Organisations & Strategy

3.1 What Is an Organisation?

View	Definition
Technical / Microeconomic	Formal social structure using resources from the environment to produce outputs; a legal entity with internal rules & social structure
Behavioural	A collection of rights, privileges, obligations & responsibilities balanced over time through conflict and resolution

3.2 Key Features of Organisations

- Hierarchical structure
- Accountability & authority in impartial decision-making
- Adherence to principle of efficiency (routines & SOPs)
- Business processes: collections of routines → collection = the firm
- Organisational politics, culture, environment

3.3 Economic Impact of IT on Organisations

Transaction Cost Theory

- Firms seek to minimise transaction costs (costs of participating in markets).
- IT lowers market transaction costs → firms can transact externally rather than growing headcount (supports outsourcing).

Agency Theory

- Firm = nexus of contracts among self-interested parties needing supervision (agency costs).
- IT reduces agency costs → firms can grow without proportional increase in managerial overhead.

3.4 Behavioural / Organisational Impact

- **IT flattens organisations:** decision-making pushed lower; fewer managers needed.
- Post-industrial orgs: authority from knowledge & competence, not just formal rank.
- **Resistance to change:** IS bound up in organisational politics; changes structure, culture & work.
- Four factors of resistance: nature of IT innovation, org structure, org culture, tasks affected.

3.5 Porter's Competitive Forces Model

Identifies five competitive forces that shape industry profitability and strategic options:

Force	Description + IS Response
Threat of New Entrants	New competitors entering the market. IS can build entry barriers (e.g., loyalty programmes, proprietary tech).
Bargaining Power of Suppliers	Suppliers' ability to raise prices. IS enables supplier switching or vendor portals (VMI).
Bargaining Power of Buyers	Customers' ability to demand lower prices. IS enables mass personalisation and switching costs.
Threat of Substitutes	Alternative products/services. IS enables differentiation and innovation.
Rivalry Among Existing Competitors	Intensity of competition. IS provides cost leadership, differentiation, or focus strategies.

3.6 IS Strategies for Competitive Forces

- **Low-cost leadership:** Use IS to produce at lower cost than competitors (e.g., Walmart).
- **Product differentiation:** Use IS to create unique products/services (e.g., Nike custom shoes).
- **Focus on market niche:** Use IS to target a specific market (e.g., Hilton customer analytics).
- **Customer & supplier intimacy:** Use IS to tighten relationships (e.g., Amazon 1-Click, JC Penney VMI).

3.7 Value Chain Model

Value Chain: Views the firm as a series of activities that each add value to products/services.

Activity Type	Examples
Primary Activities	Inbound logistics, Operations, Outbound logistics, Sales & Marketing, Service
Support Activities	Firm infrastructure, HR management, Technology development, Procurement

- **Value Web:** Collection of independent firms using IT to coordinate their value chains for a market collectively — flexible & adaptive.

3.8 Network-Based Strategies

- **Network Economics:** Value grows exponentially with number of members (Metcalf's Law). E.g., WhatsApp, Zomato.
- **Virtual Company Model:** Links people, assets & ideas across firms via network — no physical ownership needed. E.g., Li & Fung manages garment production for Guess/Levi's.
- **Business Ecosystems & Platforms:** Loosely coupled networks with keystone firms (e.g., Facebook Marketplace, Apple App Store).

■ **Exam Tip:** Porter's 5 Forces + Value Chain + IS strategy responses are high-frequency exam topics.

Chapter #5

IT Infrastructure & Emerging Technologies

5.1 Defining IT Infrastructure

IT Infrastructure: Set of physical devices and software required to operate the enterprise — a "service platform."

Infrastructure Services Include

- Computing Services
- Telecommunication Services
- Data Management Services
- Application Software Services
- Physical Facilities Management
- IT Management, Standards, Education, R&D; Services

5.2 Evolution of IT Infrastructure

Era	Description
1959+ Mainframe & Minicomputer	General-purpose centralised computing
1981+ Personal Computer	Desktop computing, individual productivity
1983+ Client/Server	Networked PCs and servers
1992+ Enterprise Computing	Integrated ERP, internet-connected
2000+ Cloud & Mobile	On-demand, anywhere access, smartphones
2010+ Big Data & Deep Learning	Massive datasets, neural networks
2020+ Generative AI & Foundation Models	LLMs, multimodal AI, ChatGPT era

5.3 Technology Drivers of Infrastructure Evolution

Moore's Law

- Processing power doubles every ~18 months; cost halves every 18 months.
- Variants: microprocessor power / computing power / price of computing.

Nanotechnology

- Uses individual atoms & molecules to build chips thousands of times smaller.

Law of Mass Digital Storage

- Amount of digital data doubles ~every year; cost of storage falls ~100% per year.

Metcalf's Law

- Value/power of a network grows **exponentially** with number of members.
- Explains "increasing returns" in social networks, messaging apps.

Declining Communications Costs & Internet

- Drastic fall in bandwidth costs → global e-commerce, cloud computing viable.

5.4 Components of IT Infrastructure

Component	Key Players / Details
Computer Hardware	Client machines (PCs, laptops, smartphones, tablets); Servers
OS Platforms	Servers: Windows Server, Unix, Linux; Clients: Windows, Android, iOS, Chrome OS

Component	Key Players / Details
Enterprise Software	SAP, Oracle (>\$500B spend 2020); IBM, Oracle middleware
DBMS	IBM DB2, Oracle, MS SQL Server, MySQL, MongoDB, Apache Hadoop
Networking / Telecom	TCP/IP standard; Cisco/Juniper hardware; AT&T, Verizon, Airtel, Jio services
Internet Platforms	Web hosting, DNS (Cloudflare, Amazon Route 53, Google Cloud DNS)
Consulting & SI	Accenture, IBM GS, Infosys, Wipro — systems integration & legacy support

5.5 Key Hardware Trends

Trend	Key Points
Mobile Digital Platform	Smartphones, tablets, wearables; BYOD forces new IT policies
Quantum Computing	Uses quantum physics — dramatic speed increases (IBM)
Virtualisation	One physical resource → multiple virtual resources; VMware; reduces hardware costs
Cloud Computing	IaaS / SaaS / PaaS; public, private, hybrid cloud; scalable, flexible
Edge Computing	Servers near data source; reduces latency; good for IoT & geographically spread users
Green Computing	Data centres = ~1.5% world energy; green practices, multicore/power-efficient processors

5.6 Cloud Computing Service Models

IaaS (Infrastructure as a Service) — Virtual machines, storage, networking. *E.g., AWS EC2.*

PaaS (Platform as a Service) — Development tools & runtime. *E.g., Google App Engine.*

SaaS (Software as a Service) — Ready-to-use applications over the internet. *E.g., Salesforce, Google Workspace.*

■ Cloud Layers Mnemonic

I P S

- IaaS — Infrastructure (raw compute)
- PaaS — Platform (build on it)
- SaaS — Software (just use it)

5.7 Key Software Trends

- **Open-source:** Linux, Apache, Mozilla Firefox; community-built, free to use.
- **Web technologies:** HTML5, Java, Python, Ruby — enable browser-based apps.
- **SOA (Service-Oriented Architecture):** Self-contained, reusable services communicate via APIs — modular software design.
- **Web Services & XML:** Universal data exchange standard across different platforms.
- **Cloud-based software:** SaaS, mashups, mobile apps replacing on-premise installs.

5.8 Challenges in Managing IT Infrastructure

Challenge	Key Consideration
Platform & infrastructure change	Scalability, flexibility; cloud vs on-premise; BYOD & MDM policies

Challenge	Key Consideration
Centralisation vs decentralisation	Should IT be managed centrally or by departments?
TCO (Total Cost of Ownership)	Hardware + software + maintenance + training + downtime — not just purchase price
Competitive Forces Model for IT	Market demand, technology vendors, rivals, standards, legacy systems pressure
■ Exam Tip: <i>Moore's Law, Metcalfe's Law, IaaS/PaaS/SaaS and TCO are must-know exam concepts for this chapter.</i>	

Chapter #6

Foundations of Business Intelligence: Databases & Information Management

6.1 Data Hierarchy

Bit	Byte	Field	Record	File	Database
-----	------	-------	--------	------	----------

6.2 Traditional File Problems vs DBMS Solutions

Problem (Traditional File)	DBMS Solution
Data redundancy & inconsistency	Centralised DB controls redundancy
Lack of flexibility	Multiple views from single DB
Program-data dependence	Logical/physical view separation
Poor data security	Centralised access control & permissions

6.3 Relational DBMS

Relational DB: Represents data as 2-D tables (rows = records/tuples; columns = fields/attributes).

Key Term	Meaning
Primary Key	Uniquely identifies each record in a table
Foreign Key	Primary key of one table used as look-up in another
SELECT	Creates subset of records meeting stated criteria (WHERE clause)
JOIN	Combines two tables to provide richer information
PROJECT	Creates subset of columns (SELECT specific columns)

Key DB Capabilities

- Data definition & data dictionary
- SQL (Structured Query Language) for querying & manipulation
- Report generation (e.g., Microsoft Access)
- Normalisation — eliminate data redundancy

6.4 Beyond Relational: NoSQL, Cloud, Distributed DBs

Type	Key Details
Non-Relational (NoSQL)	Flexible schema; designed for large distributed datasets. E.g., MongoDB, Oracle NoSQL, Amazon SimpleDB
Cloud Database	SQL/NoSQL engines hosted on cloud; pay-per-use pricing model
Distributed Database	Stored across multiple physical locations. E.g., Google Spanner
Blockchain	Distributed ledger; verified transactions; immutable records; no central authority

6.5 Big Data

3 Vs of Big Data:

- **Volume** — Petabytes, Exabytes of data
- **Variety** — Structured, semi-structured, unstructured (web, social, sensors)
- **Velocity** — Speed at which data is generated and must be processed

6.6 Business Intelligence Infrastructure

Tool	Purpose & Key Features
Data Warehouse	Stores current & historical data from many TPS; consolidated, standardised; read-only; analysis & reporting
Data Mart	Subset of data warehouse; single subject or line of business
Hadoop	Distributed parallel processing of big data across cheap computers; HDFS (storage) + MapReduce (processing) + HBase (NoSQL DB)
In-Memory Computing	Uses RAM instead of disk storage; hours → seconds. E.g., SAP HANA, Oracle DB in-Memory
Analytical Platforms	High-speed relational + non-relational hybrid platforms for very large datasets

6.7 Analytical Tools

Tool	Description
OLAP (Online Analytical Processing)	Multi-dimensional analysis — slice data by product, region, time, etc. Answers ad hoc queries rapidly.
Data Mining	Finds hidden patterns & relationships. Types: Associations, Sequences, Classification, Clustering, Forecasting
Text Mining	Extracts key elements from unstructured text; includes sentiment analysis
Web Mining	Analyses web content, structure & usage patterns

■ **Exam Tip:** Difference between Data Warehouse vs Data Mart, and the 3 Vs of Big Data are very common exam questions.

Chapter #8

Securing Information Systems

8.1 Core Definitions

Term	Definition
Security	Policies, procedures & technical measures to prevent unauthorised access, alteration, theft or damage to IS
Controls	Methods, policies & org procedures that ensure safety of assets, accuracy of records, adherence to management standards

8.2 Why IS Are Vulnerable

Sources of Vulnerability

- Network accessibility — open to anyone
- Hardware problems (breakdowns, config errors)
- Software bugs and installation errors
- Disasters (natural or man-made)
- Use of networks outside firm control
- Loss/theft of portable devices
- Wireless/Bluetooth networks — easy to scan (war driving, rogue APs)

8.3 Types of Threats

Threat	Description	Example
Malware	Viruses, worms, Trojan horses, spyware	ZeuS Trojan — keylogger capturing banking credentials
Ransomware	Encrypts data; demands payment	Sodinokibi — 22 Texas cities (2019)
SQL Injection	Rogue SQL query via input field attacks DB	Attacker inserts malicious SQL in a web form
DoS / DDoS	Floods server; denies service to legit users	Mirai botnet (2016) — took down Twitter, Netflix etc.
Phishing / Spear Phishing	Fake websites/emails to steal credentials	Evil twins (bogus Wi-Fi); Pharming (redirect to bogus page)
Identity Theft	Stealing personal info for fraud	\$16.9B loss in US (2019)
Click Fraud	Fraudulent clicks on pay-per-click ads	Serious issue for Google AdSense
Cyberterrorism / Cyberwarfare	State-sponsored attacks to disrupt nations	Nation-state hacking of critical infrastructure
Spoofing & Sniffing	Impersonating IP addresses; intercepting data packets	Packet sniffer on unsecured Wi-Fi
Social Engineering	Manipulating people to divulge info	Pretexting calls to IT helpdesk

8.4 Organisational Framework for Security & Control

- **IS Controls:** Input, processing & output controls; application controls; general controls.
- **Risk Assessment:** Determine value of assets, probability & impact of threats, and vulnerabilities.
- **Security Policy:** Defines acceptable use; ranks information risks; establishes procedures.
- **Disaster Recovery Planning (DRP):** Restores disrupted IT services after a disaster.

- **Business Continuity Planning (BCP):** Focuses on restoring business operations after crisis.
- **Security Auditing:** Reviews controls, tests for weaknesses, checks compliance.

8.5 Legal & Regulatory Requirements

Law / Regulation	Scope
HIPAA	Medical data security & privacy (US healthcare)
Gramm-Leach-Bliley Act	Security & confidentiality of customer financial data
Sarbanes-Oxley Act	Accuracy & integrity of financial information; corporate accountability

8.6 Tools & Technologies for Safeguarding IS

Tool / Tech	How It Works
Identity Management	Automates tracking users & privileges; authenticates identities
Authentication	Passwords, Tokens, Smart Cards, Biometrics, Two-Factor Authentication (2FA)
Firewall	Hardware + software barrier; packet filtering, stateful inspection, NAT, proxy filtering
Intrusion Detection System (IDS)	Monitors hot spots on networks; detects & alerts on suspicious activity
Anti-Malware / Anti-Spyware	Checks for & removes malware; requires continual updates
UTM (Unified Threat Management)	Combines firewall, IDS, anti-malware in one appliance
Encryption	Transforms plain text → cipher text. SSL for connections; S-HTTP for individual messages
Digital Certificates (PKI)	Establishes identity of users/assets for online transactions; issued by Certificate Authorities
Blockchain Security	Distributed, encrypted, immutable ledger for secure transactions

8.7 Wireless & Cloud/Mobile Security

Wireless Standards

- **WEP:** Weak static encryption — easily cracked; use with VPN only.
- **WPA2:** Stronger, continually changing longer keys — standard today.
- **WPA3:** Most recent, strongest encryption — recommended.

Cloud Security

- Responsibility for security stays with the company owning the data.
- Firms must verify: data location, legal privacy compliance, data segregation, SLA terms.

Mobile Security

- MDM (Mobile Device Management): authorisation, inventory, remote lock/wipe, encryption.
- Software for segregating corporate vs personal data on BYOD devices.

■ **Exam Tip:** Know the types of malware, the regulatory acts (HIPAA, SOX), and all safeguarding technologies.

Chapter #9

Achieving Operational Excellence & Customer Intimacy: Enterprise Applications

9.1 Enterprise Systems (ERP)

ERP (Enterprise Resource Planning): Suite of integrated software modules sharing a common central database.

- Information entered once is immediately available firm-wide.
- Built around pre-defined best-practice business processes (Finance, HR, Manufacturing, Sales).
- Firms select functions → map their processes → use configuration tables (SAP has 3,000+) for customisation.

Business Value of ERP

- ↑ Operational efficiency
- Firm-wide information for decision making
- ↑ Speed of response to customers
- Analytical tools for performance evaluation

■ Case Example — Alcoa using Oracle ERP:

- Eliminated redundant processes across 31 countries, 200+ locations
- Accounts payable processing ↓ by 89%
- Worldwide costs ↓ by ~20%

9.2 Supply Chain Management (SCM)

Supply Chain: Network of organisations & processes for procuring, transforming & distributing products.

Supply Chain Type	Description
Upstream	Firm → Suppliers (raw materials, components)
Internal	Within the firm (production, assembly, operations)
Downstream	Firm → Distributors → Customers

Key SCM Concepts

- **Just-In-Time (JIT):** Components arrive exactly when needed; no excess inventory.
- **Safety Stock:** Buffer inventory for supply chain inflexibility.
- **Bullwhip Effect:** Small rise in customer demand → amplified overstocking up the chain. E.g., P&G; Pampers excess.

SCM Software

- **Supply Chain Planning:** Models supply chain, enables demand planning, optimises sourcing/inventory/transport.
- **Supply Chain Execution:** Manages flow through distribution centres & warehouses.

Push vs Pull Models

Model	Key Feature
Push (Build-to-Stock)	Schedules based on demand forecasts; risk of over/under production
Pull (Demand-Driven)	Customer orders trigger supply chain events; internet-enabled; reduces waste

Business Value of SCM

- Match supply to demand
- Reduce inventory levels
- Improve delivery service

- Speed product time to market
- Increase sales; reduce total supply chain costs (~75% of operating budget)

9.3 Customer Relationship Management (CRM)

CRM: Systems to capture, consolidate, analyse & distribute customer data firm-wide for a single enterprise view of the customer.

CRM Software Modules

- **SFA (Sales Force Automation):** Contact info, quote generation, pipeline management.
- **Customer Service:** Assign & manage service requests; web self-service.
- **Marketing Automation:** Track direct marketing; cross-selling.
- **PRM (Partner Relationship Management):** Lead gen, pricing, promotions for channel partners.
- **ERM (Employee Relationship Management):** Objectives, performance, training.

Operational vs Analytical CRM

Type	Focus
Operational CRM	Customer-facing: SFA, call centre, marketing automation — day-to-day interactions
Analytical CRM	Data warehouse-based: OLAP & data mining on customer data; calculates CLTV (Customer Lifetime Value)

Business Value of CRM

- ↑ Customer satisfaction & retention
- ↓ Direct marketing costs
- ↑ Sales revenue
- ↓ Customer acquisition cost
- Reduces churn rate (customers leaving)

9.4 Enterprise Application Challenges

- Expensive to purchase & implement; long development times
- Technology changes require constant updates
- Business process re-engineering often required
- High switching costs & vendor dependence
- Data standardisation & cleansing across legacy systems

9.5 Next-Generation Enterprise Applications

- **Cloud-based ERP/CRM:** Flexible, scalable SaaS delivery.
- **Mobile platform versions:** Access on smartphones/tablets.
- **Social CRM:** Integrates social media; monitors social activity; social analytics.
- **BI integration:** Built-in flexible reporting, "what-if" scenarios, digital dashboards, AI/ML.

■ **Exam Tip:** Know the Bullwhip Effect, Push vs Pull models, Operational vs Analytical CRM, and CLTV.

Chapter #10

E-Commerce: Digital Markets, Digital Goods

10.1 E-Commerce Overview

E-Commerce: Use of Internet & Web to transact business. Began 1995; grew exponentially.

- The "New" E-commerce: **Social** (networks), **Mobile** (smartphones), **Local** (geo-targeted).
- Shift from desktop to smartphone; from "eyeballs" to "conversations" (conversational commerce).

10.2 Why E-Commerce Is Different (8 Unique Features)

UGRI·IPP·S mnemonic (or just learn the 8):

1. **Ubiquity** — Anytime, anywhere; reduces transaction costs
2. **Global Reach** — Crosses cultural/national boundaries
3. **Universal Standards** — One set of Internet technology standards
4. **Richness** — Video, audio, text messages
5. **Interactivity** — Two-way communication
6. **Information Density** — Greater price/cost transparency; price discrimination
7. **Personalisation / Customisation** — Modified messages and goods
8. **Social Technology** — User content generation; social networks

10.3 Key E-Commerce Concepts

Concept	Meaning
Information Asymmetry	One party has more information than the other; Internet reduces it
Menu Costs	Cost of changing prices; near-zero online
Dynamic Pricing	Prices change based on demand/supply in real-time
Switching Costs	Cost for customer to move to a competitor; lock-in strategy
Disintermediation	Elimination of intermediaries; manufacturer sells directly to consumer

10.4 Types of E-Commerce

Type	Description & Example
B2C (Business-to-Consumer)	Amazon, Flipkart, Myntra
B2B (Business-to-Business)	Indiamart — B2B estimated at \$14.5T in US (2020); e-commerce portion \$6.7T
C2C (Consumer-to-Consumer)	OLX, Quikr
M-Commerce (Mobile)	~45% of all e-commerce (2020); growing >20% p.a.

10.5 E-Commerce Business Models

Model	Description & Example
Portal	Entry point with search & content. E.g., Google, Yahoo
E-tailer	Online retail store. E.g., Amazon, Flipkart
Content Provider	Digital content subscription. E.g., Netflix, Spotify
Transaction Broker	Earns fee per transaction. E.g., Booking.com, MakeMyTrip
Market Creator	Creates marketplace. E.g., eBay, OLX

Model	Description & Example
Service Provider	Provides services online. E.g., Dropbox, Google Drive
Community Provider	Shared interest platform. E.g., Facebook, LinkedIn

10.6 E-Commerce Revenue Models

Revenue Model	How It Earns	Example
Advertising	Display paid ads to users	Google AdSense, banner ads
Sales	Direct sale of goods/services	Amazon, Flipkart
Subscription	Recurring fee for access	Netflix, Spotify
Free/Freemium	Basic free; premium paid	LinkedIn, Dropbox
Transaction Fee	Fee per enabled transaction	PayPal, booking platforms
Affiliate	Commission for steering traffic	Affiliate marketing links

10.7 E-Commerce Marketing Transformation

Marketing Formats

- Search (Google Ads)
- Display ads (banners)
- Video & rich media
- Email marketing
- Long-tail marketing

Targeting Techniques

- **Behavioural Targeting:** Tracks online behaviour across sites to serve relevant ads.
- **Programmatic Ad Buying:** Automated real-time bidding for ad inventory.
- **Native Advertising:** Ads that blend with content style.

Social Commerce Concepts

- **Social Graph:** Map of online social relationships — used to target ads and recommendations.
- **Wisdom of Crowds:** Large groups make better decisions than small committees.
- **Crowdsourcing:** Firms solicit solutions from a large community.

10.8 B2B E-Commerce Methods

Method	Description	Relationship Model
EDI (Electronic Data Interchange)	Computer-to-computer exchange of standard transaction documents (invoices, POs)	1-to-1 continuous replenishment
Private Industrial Networks	Secure website linking large firm with specific suppliers/partners	1-to-Many; anchor firm centred
Net Marketplaces (e-hubs)	Digital marketplace for many buyers & sellers; vertical or horizontal focus	Many-to-Many shared platform
Exchanges	Independently owned 3rd-party Net marketplace for spot purchasing	Many-to-Many; short-term transactions

10.9 M-Commerce & Location Services

- M-commerce = 45% of e-commerce (2020); fastest growing.

- **Location-Based Services:** GPS map + geo-social (friends) + geo-advertising (nearby shops) + geo-info (property prices).

Mobile Payment Types

- **NFC (Near Field Communication):** Apple Pay, Google Pay, Samsung Pay — tokenised credit cards.
- **QR Code:** UPI-based payments (dominant in India).
- **P2P (Peer-to-Peer):** UPI apps (PhonePe, GPay) using mobile numbers.

■ **Exam Tip:** The 8 unique features of e-commerce, B2B methods (EDI vs Net Marketplace), and revenue models are exam favourites.

Chapter #11

Managing Knowledge & Artificial Intelligence

11.1 Role of Knowledge Management

Information Economy: Production & distribution of information and knowledge is a major source of wealth.

- Substantial part of a firm's stock market value is **intangible assets**: knowledge, brands, reputation, processes.
- Well-executed knowledge-based projects can produce extraordinary ROI.

11.2 Dimensions of Knowledge

Dimension	Key Points
Tacit Knowledge	Hard to articulate; resides in people; learned through experience; e.g., "know-how"
Explicit Knowledge	Codified, documented; easy to share; e.g., manuals, reports, databases
Knowledge as Asset	Knowing how to do things effectively = source of profit & competitive advantage
Knowledge has Location	Resides in individuals, groups, or organisational processes
Knowledge is Situational	Relevant only in specific contexts

11.3 Knowledge Management Value Chain

Knowledge Acquisition	Knowledge Storage	Knowledge Dissemination	Knowledge Application
Stage	Activities	Tools	
Acquisition	Document tacit/explicit knowledge; track TPS data; develop expert networks	Documents, reports, email archives, expert directories	
Storage	Store captured knowledge for future retrieval	Databases, document management systems	
Dissemination	Share knowledge across org; manage information overload	Portals, wikis, email, IM, search engines, collaboration tools	
Application	Apply knowledge to create value	New business practices, new products, new markets	

11.4 Types of Knowledge Management Systems

System Type	Description
Enterprise-Wide KMS	General-purpose firm-wide effort to collect, store, distribute & apply digital content and knowledge
Knowledge Work Systems (KWS)	Specialised for engineers, scientists, knowledge workers to discover & create knowledge
Intelligent Techniques	Diverse: data mining, expert systems, ML — for discovering & distilling knowledge, finding optimal solutions

11.5 Enterprise Content Management (ECM) & LMS

Three Types of Knowledge in an Enterprise

- **Structured:** Reports, presentations, formal rules (~20%).
- **Semi-structured:** Emails, videos (~30%).

- **Unstructured / Tacit:** Tacit knowledge, informal (~50%). — 80% of org content is semi- or unstructured!

Learning Management Systems (LMS)

- Manages, delivers, tracks & assesses employee learning
- Supports web-based classes, online forums, certifications
- Automates course selection & administration
- Enables MOOCs (Massively Open Online Courses)

11.6 What Is Artificial Intelligence (AI)?

AI: Computer programmes capable of performing complex tasks that normally require human intelligence — interpreting CT scans, playing chess, recognising faces.

Drivers of AI Evolution

- Big Data (Internet, IoT, social media)
- Drastic reduction in processing costs
- Refinement of algorithms by researchers
- Massive investment by business & governments

Alan Turing (1950): AI = a programme a human cannot distinguish from a human in conversation (Turing Test).

11.7 Major AI Techniques

Technique	Description & Example
Expert Systems	Captures expert knowledge as IF-THEN rules; ~20% of AI systems. E.g., medical diagnosis tools
Machine Learning (ML)	Systems learn by recognising patterns in large datasets (neural networks, genetic algorithms)
NLP (Natural Language Processing)	Computer understands & analyses natural human language. E.g., chatbots, Alexa, Siri
Computer Vision	Computers emulate human vision to extract info from images. E.g., facial recognition, medical imaging
Robotics	Design & operation of movable machines; computer control & sensory feedback
Intelligent Agents	Software that works in background without direct human intervention to carry out tasks

11.8 Machine Learning Types

Type	Data Used	Tasks & Examples
Supervised Learning	Labelled data (input-output pairs)	Classification & Regression. E.g., SPAM detection, image classification, sentiment analysis
Unsupervised Learning	Unlabelled data	Clustering, Associations, Anomaly detection. E.g., customer segmentation, fraud detection
Reinforcement Learning	Environment interactions	Sequential decision-making. E.g., Chess/Go AI, autonomous cars, ChatGPT alignment

11.9 Neural Networks & Genetic Algorithms

Neural Networks

- Interconnected units (neurons) that detect patterns in large data quantities.
- Finding pathways — e.g., diagnosing epilepsy/Alzheimer's; credit card fraud detection.

Genetic Algorithms

- Examines large number of alternative solutions → finds optimal solution.
- Inspired by evolutionary biology (mutation, selection, crossover).
- E.g., optimise design of jet turbine engines.

11.10 Knowledge Work Systems (KWS)

- **Knowledge Workers:** Researchers, designers, architects, scientists, engineers.
- **CAD/CAM:** Computer-aided design; 3D printing.
- **Virtual Reality:** Simulate real-life environments.
- **AR (Augmented Reality):** Overlays digital data/images onto physical world.

■ **Exam Tip:** Know the 3 ML types, Neural Networks vs Genetic Algorithms, and all AI techniques with examples.

Chapter #12

Enhancing Decision Making

12.1 Types of Decisions

Type	Characteristics	Who Makes It
Unstructured	Judgment, evaluation & insight required; no clear procedure	Senior executives (C-level)
Semi-Structured	Partly clear-cut; combination of procedure & judgment	Middle managers
Structured	Repetitive, routine; definite procedure; can be automated	Operational managers & staff

12.2 Decision-Making Process (Simon, 1960)

INTELLIGENCE (Identify problems)	DESIGN (Explore solutions)	CHOICE (Select alternative)	IMPLEMENTATION (Execute & monitor)
----------------------------------	----------------------------	-----------------------------	------------------------------------

12.3 Managerial Roles (Mintzberg's 10 Roles)

Category	Role	IS Support?
Interpersonal	Figurehead, Leader, Liaison	Limited — mainly relationship-based
Informational	Nerve centre, Disseminator, Spokesperson	Strong — data, reports, dashboards
Decisional	Entrepreneur, Disturbance handler, Resource allocator, Negotiator	Partial — DSS, BI tools help

■ **Note:** IS can support SOME managerial roles, but NOT all (especially interpersonal/decisional judgment roles).

12.4 Why IT Investments Don't Always Work

- **Information Quality:** Poor quality data → poor decisions (GIGO — Garbage In, Garbage Out).
- **Management Filters:** Selective attention & cognitive biases; managers reject info inconsistent with prior beliefs.
- **Organisational Inertia & Politics:** Strong internal forces resist change-requiring decisions.

12.5 High-Velocity Automated Decision Making

Enabled by algorithms that define precise steps for structured decisions — humans removed from loop.

- Example: High-speed computer trading (nanosecond execution) — Intelligence, Design, Choice & Implementation all done by software.
- Requires safeguards & regulation to prevent runaway automated decisions.

12.6 Business Intelligence (BI)

Business Intelligence (BI): Infrastructure for collecting, storing & analysing data produced by the business.

BI Component	Description
BI Infrastructure	Databases, data warehouses, data marts, Hadoop, analytic platforms
Business Analytics	Tools for analysing data: OLAP, statistics, models, data mining
BI Vendors	SAP, IBM Cognos, Oracle, Microsoft BI, Tableau etc.

6 Elements of the BI Environment

- Data from business environment
- BI infrastructure
- Business analytics toolset
- Managerial users & methods
- Delivery platform (MIS, DSS, ESS)
- User interface / data visualisation tools

12.7 BI Analytic Capabilities

Capability	Description
Pre-defined Production Reports	Standard scheduled reports (e.g., monthly sales)
Parameterised Reports (Pivot Tables)	User adjusts parameters for customised output
Dashboards / Scorecards	Real-time KPI visualisations
Ad-hoc Query / Search	User creates own queries on the fly
Drill Down	Explore summary → detail level
Forecasts / Scenarios / What-If	Model future outcomes; sensitivity analysis

12.8 Predictive Analytics

Predictive Analytics: Uses statistical analysis, data mining & historical data to estimate probability of future events.

- Works well when: large data sets, repetitive & consistent behaviour, stable environment.
- Does NOT work well in dynamic environments (e.g., stock market with constantly changing variables).
- Examples: credit scoring, response prediction for direct marketing.

12.9 Big Data Analytics & Location Analytics

- **Big Data Analytics:** Analysing massive datasets from social media, in-store, online — enables real-time personalisation.
- **Smart Cities:** Public records + sensor data + smartphones → evaluate effect of policy changes.
- **Operational Intelligence:** Business Activity Monitoring; IoT sensor data streams → real-time analysis.
- **Location Analytics:** Business insights from geographical component of data — mobile ad targeting.
- **GIS (Geographic Information Systems):** Visualise and map problems — disaster response, city planning.

12.10 Decision Support for Different Levels

Management Level	System & Key Features
Operational / Middle	MIS — structured/semi-structured decisions; exception reports; predictive maintenance
Semi-Structured Decisions	DSS — what-if analysis, sensitivity analysis, backward analysis, OLAP (pivot tables)
Senior / C-Level	ESS — Balanced Scorecard (Financial, Process, Customer, Learning & Growth dimensions); KPIs; BPM

Balanced Scorecard

4 Dimensions: **Financial** | **Business Process** | **Customer** | **Learning & Growth**

- Each dimension measured by KPIs (Key Performance Indicators)
- BPM (Business Performance Management) translates strategy into operational targets

■ **Exam Tip:** Simon's 4-step decision process, Mintzberg's 10 roles, BI capabilities, and Balanced Scorecard are classic exam questions.

Chapter #13

Building Information Systems

13.1 Systems Development & Organisational Change

Building a new IS is a form of **planned organisational change** — even if it affects only one functional unit.

Degrees of Organisational Change (Low → High Risk & Reward)

Automation	Rationalisation of Procedures	Business Process Redesign	Paradigm Shift
------------	-------------------------------	---------------------------	----------------

Type	Description
Automation	Replaces manual tasks; increases efficiency. E.g., automated payroll.
Rationalisation of Procedures	Streamlines SOPs; TQM, Six Sigma programmes.
Business Process Redesign (BPR)	Analyses, simplifies & redesigns processes; eliminates waste & repetition.
Paradigm Shift	Rethinks the entire business model; changes the nature of the organisation. Highest risk.

13.2 Business Process Management (BPM)

BPM: Tools & methodologies to analyse, design & optimise business processes.

5 Steps of BPM

1. Identify Processes for Change	2. Analyse Existing Processes	3. Design New Process	4. Implement New Process	5. Continuous Measurement
----------------------------------	-------------------------------	-----------------------	--------------------------	---------------------------

13.3 Core Activities in Systems Development

Systems Analysis	Systems Design	Programming	Testing	Conversion	Production & Maintenance
------------------	----------------	-------------	---------	------------	--------------------------

Activity	Key Points
Systems Analysis	Define problem, identify causes, specify solutions, identify information requirements; produce feasibility study & systems proposal
Systems Design	Describes specifications (managerial, org & tech). User involvement critical — insufficient user involvement is major cause of failure
Programming	Translate design into software code
Testing	Unit testing, system testing, acceptance testing; test plan required
Conversion	Switch from old to new system (see 4 strategies below)
Production & Maintenance	Operate and monitor; revisions/modifications as needed

4 Conversion Strategies

Strategy	Description
Parallel	Old & new systems run simultaneously; safest; most expensive
Direct Cutover	Old system turned off when new turned on; risky; cheapest
Pilot Study	New system tested in one part of org first
Phased Approach	New system introduced in stages/modules

13.4 Modelling & Design Methodologies

Structured Methodology

- Step-by-step, top-down approach (abstract → detail).
- Focus on modelling processes — separates data from process.
- **Tool:** DFD (Data Flow Diagram) — shows data flows, processes, data stores & external entities.

Object-Oriented Development

- Basic unit = **Object** (combines data + processes; data accessible only via associated operations).
- **Class & Inheritance:** Objects belong to a class; can inherit from parent classes — promotes reuse.
- More iterative & incremental than structured; systems analysis identifies interactions.
- Objects are reusable → potentially reduces time & cost of development.

13.5 System Building Methods

Method	Key Characteristics
Traditional Systems Life Cycle (Waterfall)	Oldest; phased (one stage completes before next); formal division of labour; costly, time-consuming, inflexible; still used for large complex systems
Prototyping	Build experimental system rapidly & cheaply; user evaluates prototype; iterative refinement; good for uncertain requirements; risk of glossing over essential steps
Application Software Packages	Buy off-the-shelf; use RFP (Request for Proposal) to evaluate vendors; customisation features; saves time & money
Outsourcing (domestic/offshore)	External vendors design & build; SaaS & cloud providers; advantage: flexibility; disadvantage: hidden costs, proprietary process exposure
RAD (Rapid Application Development)	Workable system in short time; often uses JAD (Joint Application Design) for requirements
Agile Development	Rapid delivery via small sub-projects; working software quickly; iterative
DevOps	Agile + continuous deployment; bridges development & operations
Low-Code / No-Code	Visual modelling; minimal hand-coding; limited functionality; risk of ungoverned apps

13.6 Mobile Application Development

- **Mobile Websites:** Optimised website accessed via mobile browser.
- **Mobile Web Apps:** Web-based apps mimicking native app experience.
- **Native Apps:** Built specifically for iOS/Android; best performance; uses device hardware.
- **Special Requirements:** Smaller screens, multitouch, conserve memory & battery.
- **Responsive Web Design:** Layout changes automatically based on device screen size.

13.7 IS Leadership Challenges Summary

Challenge	Example / Consideration
New or Customise?	Buy SaaS (Clio for law firm) OR build proprietary (Goldman Sachs Marquee trading platform)
How much to spend?	Business case ROI analysis — \$50k robot saving \$20k/yr = good; saving \$2k = bad
What capabilities?	Single dept (hospital radiology) vs enterprise-wide (full ERP); primary partners vs all stakeholders
Scalable & Flexible?	Netflix AWS auto-scales; Zara needs flexible inventory logic every 2 weeks
Security level?	Nuclear plant = biometric + air-gapped; local café = simple password
Technology Roadmap?	HDFC 5-year: cloud → AI chatbots → blockchain → branchless banking

■ IS Leadership Decision Framework

B-SCSR (Build, Spend, Capabilities, Scalability, Security, Roadmap)

- B — Build or Buy?
- S — Spend (ROI)?
- C — Capabilities needed?
- S — Scalability & Flexibility?
- S — Security level?
- R — Roadmap for evolution?

■ **Exam Tip:** Waterfall vs Agile, 4 conversion strategies, and the 5-step BPM process are most likely to appear in the exam.

QUICK REVISION

Key Formulas, Laws & Mnemonics

Laws & Principles

Law / Principle	Statement
Moore's Law	Processing power doubles / cost halves every ~18 months
Metcalfe's Law	Network value grows exponentially with number of members
Law of Mass Digital Storage	Amount of digital data doubles yearly; storage cost falls ~100% p.a.
Bullwhip Effect	Small demand changes amplify to large swings up the supply chain
Turing Test (1950)	AI = computer indistinguishable from human in conversation

Master Mnemonic Sheet

6 IS Objectives: ON-DISC (Operational Excellence, New products, Decision making, Intimacy, Survival, Competitive advantage)

Simon's Decision Making: I-D-C-I (Intelligence, Design, Choice, Implementation)

Cloud Layers: IPS (IaaS → PaaS → SaaS)

Big Data: 3Vs (Volume, Variety, Velocity)

Knowledge Value Chain: A-S-D-A (Acquire, Store, Disseminate, Apply)

BPM 5 Steps: I-A-D-I-M (Identify, Analyse, Design, Implement, Measure)

Conversion Strategies: P-D-P-P (Parallel, Direct, Pilot, Phased)

IS Leadership: B-SCSR (Build/Buy, Spend, Capabilities, Scalability, Security, Roadmap)

EXAM CRAM

High-Yield Fast Facts

Topic	Must-Know Fact
IS vs IT	IT = hardware+software; IS = IT + people + processes + organisation
SocioTechnical	Optimise both social & technical systems — tech alone ≠ productivity
Digital Firm	Digitally mediated relationships, processes & assets; time/space shifting
Porter's 5 Forces	Rivals, New Entrants, Substitutes, Buyer Power, Supplier Power
Value Chain	Primary activities + Support activities; value web = network of firms
TPS vs MIS vs DSS vs ESS	Operational → Middle mgmt → Senior semi-structured → C-level unstructured
Moore's Law	Doubles every 18 months
Metcalfe's Law	Network value grows exponentially
Cloud IaaS/PaaS/SaaS	Raw compute / Build platform / Ready software
Big Data 3Vs	Volume, Variety, Velocity
RDBMS operations	SELECT (filter rows), JOIN (combine tables), PROJECT (select columns)
Blockchain	Distributed, immutable, encrypted ledger; no central authority
Bullwhip Effect	Demand fluctuations amplify backward in supply chain
Push vs Pull SCM	Push = forecast-based; Pull = customer-order triggered
CRM CLTV	Customer Lifetime Value — key metric in Analytical CRM
Phishing types	Spear phishing (trusted source), Evil twin (bogus Wi-Fi), Pharming (redirect)
Balanced Scorecard	4 dimensions: Financial, Process, Customer, Learning & Growth
Simon's 4 steps	Intelligence → Design → Choice → Implementation
Mintzberg roles	3 categories: Interpersonal, Informational, Decisional (10 total)
ML types	Supervised (labelled), Unsupervised (unlabelled), Reinforcement (trial & error)
Expert System	IF-THEN rules; ~20% of AI systems
Tacit vs Explicit	Tacit = know-how (hard to articulate); Explicit = codified documents
Knowledge Value Chain	Acquire → Store → Disseminate → Apply
Waterfall	Sequential phases; each completes before next; for large complex systems
Agile / DevOps	Iterative, rapid delivery; DevOps extends agile to continuous deployment
Prototyping	Quick experimental model; iterative with users; good for uncertain requirements
Conversion strategies	Parallel (safest), Direct (riskiest), Pilot, Phased
E-comm 8 features	Ubiquity, Global reach, Universal standards, Richness, Interactivity, Info density, Personalisation, Social
EDI	Computer-to-computer exchange of standard transaction documents
Disintermediation	Elimination of intermediaries in supply/distribution
HIPAA / SOX / GLB	Health privacy / Financial reporting integrity / Financial data security